

Aniket Masaye

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Profile

AI Developer and Data Scientist with a strong background in Physics and Mathematics. Expertise in building scalable ML and AI systems and optimizing API operations. Proven track record of improving multimedia experiences with ML techniques, deploying financial models, and improving hiring processes with AI driven automation. Passionate about solving complex data challenges and driving business impact.

Education

Indian Institute of Science Education and Research, Thiruvananthapuram, June 2017 – July 2022
Integrated BS-MS in Physics

- GPA: 8.0/10.0
- Thesis: Search for mono-leptonic decay of the $Z \rightarrow N_R N_R$ in $U(1)$ leptophobic mode. Built a Boosted Decision Trees model to separate signal and background for this subatomic process.
- **Coursework:** Statistical Mechanics, Real Analysis, Statistics and Probability, Mathematical methods and Programming Languages.

Experience

AI engineer, Slated Tech LLC – New York, US (Remote) June 2025 - Present

Slated.ai | *AI investment research platform transforming unstructured data into insights*

- Architected AI-powered review analysis system using LLMs and MongoDB to automatically categorize and summarize customer/employee reviews from financial platforms, reducing manual analysis time by 90% while generating actionable insights for publicly traded companies
- Built enterprise financial document pipeline processing 10,000+ investor presentations with parallel processing and robust error handling, improving document retrieval efficiency by 75% for investment analysis workflows

Data Scientist, Interactly Solutions – Hyderabad, IN Dec 2023 – June 2025

Analytos.ai | *AI-powered system for automated HTS code classification.*

- Developed AI-powered HTS classification system using FastAPI, MongoDB fuzzy search, and Pinecone vector search for semantic similarity of product descriptions.
- Implemented a hybrid HTS search strategy combining hierarchical vector search and fuzzy text search with confidence scoring and fallback mechanisms.
- Built invoice processing pipeline to extract product details and classify them with corresponding HTS codes, integrating LangChain and OpenAI models for LLM validation.
- Engineered real-time WebSocket chat interface for dynamic HTS code lookup and queries.
- Designed and maintained Prometheus metrics & Grafana dashboards to monitor request rates, tool usage, token consumption, and error metrics.
- Automated HTS data ingestion pipelines using Python scripts for uploading structured data to MongoDB and Pinecone vector database.
- Ensured system observability and reliability with structured logging, health checks, retry logic, and rate limiting safeguards.
- Deployed system using Docker & Docker Compose, enabling modular containerized environments for application, monitoring, and visualization components.

Interviewproo.ai | *AI-Powered Interview Preparation Tool*

- Built an intelligent mock interview system with FastAPI and LangChain, capable of generating personalized questions based on resumes and job descriptions.
- Designed LLM-driven agents for interactive dialogue flow and dynamic evaluation of user responses.
- Integrated analytics layer to quantify candidate performance using NLP techniques and keyword matching.

- Structured reusable modules for logging, model configuration, and NLP processing to improve scalability and maintainability.

Recruiter.ai | *AI-First Recruitment Assistant Platform*

- Developed a RAG-based AI Agent to query structured candidate data and rank applicants based on job descriptions, significantly improving hiring accuracy and decision-making.
- Engineered an AI-powered scoring system for virtual interviews using LLMs to evaluate candidates on communication skills, grammar, and articulation in real-time.
- Implemented a cheating detection system using facial recognition and iris tracking to enable real-time fraud detection, effectively reducing cheating incidents during interviews.
- Designed and deployed an automated virtual interviewer that simulates real-time technical job interviews through FastAPI, LangChain, Whisper, and ElevenLabs integrations.
- Built modular service layers to manage LLM calls, Redis queues, and Sheets-based reporting pipelines, ensuring scalable and observable interview processes.

Personaliz.ai | *AI driven platform for personalized audio, video, and speech services*

- Developed audio pitch normalization and voice cloning enhancements using custom preprocessing and pitch-shifting techniques to ensure consistent AI-generated voice output.
- Developed a speaker similarity analysis module using MFCCs and statistical voice profiling to identify voice match percentage across datasets.
- Integrated OpenAI Whisper models with transfer learning for real-time speech-to-text transcription API featuring word-level timestamps, boosting transcription accuracy across noisy datasets.
- Implemented VTT subtitle generation and video image overlay endpoints to automate personalized video creation workflows for e-commerce and content platforms.
- Refactored the API backend using FastAPI with modern pydantic models, improving request validation, modularity, and API throughput.
- Optimized AWS RDS connections using SQLAlchemy pooling strategies, enhancing scalability and reliability in concurrent database operations.
- Implemented FILM (Feature-wise Linear Modulation) via TensorFlow Hub to generate smooth intermediate video frames for low-FPS content, reducing visual jitter.
- Created bulk CSV processing workflows to generate personalized audio/video content at scale, supporting marketing and user engagement campaigns.

Data Science Intern, Interactly Solutions – Hyderabad, IN

Aug 2023 – Nov 2023

- Automated video manipulation workflows using FFmpeg, improving processing speed by 30%.
- Conducted advanced research on lip-syncing techniques and implemented LipGAN to generate high-quality lip-sync videos.
- Designed an automated video processing pipeline leveraging AWS S3 for storage and Postman for API testing, reducing manual intervention.

Data Science Intern, AI Variant – Pune, IN

Feb 2023 – Aug 2023

- Developed and deployed advanced machine learning models for financial predictions, significantly improving the accuracy of stock market forecasts.
- Led the integration of AI/ML algorithms into financial data systems, optimizing asset management processes and enhancing data-driven decision-making capabilities.
- Developed and hosted Django web application to deploy and showcase the ensemble model (with LSTM and Holt-Winters method), offering users an interactive platform.

Projects

Stock Market HFT

Feb 2025 - Present

- Designing and Developing a sub-second level prediction algorithm for stocks. Also using time-based transformers to help in long term predictions.
- Tools Used: C++, Transformers, Statistics

Counting Bird Calls from Forest Recordings

[github.com](#)

- Trained a model using PyTorch to count bird calls in forest recordings, achieving a 90% accuracy rate in audio recognition.
- Tools Used: Deep Learning, TorchAudio

Sentiment Analysis on Tweets

[github.com](#)

- Developed a Naive Bayes model to predict the sentiment of tweets related to Elon Musk, enhancing social media analytics capabilities.
- Tools Used: NLP, Naive-Bayes Classifier

Skills

Languages: Python, C++ , Java, JavaScript, SQL

Databases: MySQL, PostgreSQL, MongoDB, OracleDB

ML/AI Frameworks: TensorFlow, PyTorch, Scikit-Learn, Llama-Index, LangChain, LangGraph

Data Processing & Visualization: Pandas, NumPy, Matplotlib

Cloud & API Development: AWS (S3, RDS, EC2, SageMaker), Azure OpenAI, FastAPI, Flask

AI Specialties: Transfer Learning, Transformer Models, NLP, Time-Series Analysis, Financial Forecasting